

MATHERA HISTORY NETWORK

100₁₂ Concepts · seven technologies of mathematical thought · twelve mathematical worlds

A revised design brief: from a generic history timeline to a network of concepts, people, artifacts, transmission, controversy and technological epochs.

VERSION 2 · OCTOBER 2026

1. Changelog - how the idea evolved

Stage	Decision
V0 · Skill Tree	Mathera began as seven learner Eras, 87 Branches and 1,132 Skills: a route through mathematics.
V1 · Concept Codex	We separated repeated practice from underlying capability. The Skill Tree stays deliberately repetitive; $144 = 100_{12}$ cross-curricular Concepts aggregate mastery across branches and years.
V1.1 · RPG rarity	Concepts briefly received rarity-like colors and maximum tiers. This was visually fun but semantically wrong: Addition is not 'common loot' and Determinant is not 'rare' in the useful sense.
V1.2 · Historical depth	Rarity became historical depth. Concepts could have different numbers of genuinely distinct forms, with dates showing how an idea accumulated abstraction over time.
V1.3 · Seven historical periods	We first used Origins, Classical, Medieval, Renaissance, Enlightenment, Industrial and Modern. Useful aesthetically, but too close to a generic European-history periodization.
V2 · Technology epochs	The seven Eras are now defined by changes in the technology of doing, storing, transmitting and automating mathematics: MARK, WRITE, ALGORITHM, PRINT, MECHANIZE, COMPUTE, INTELLIGENCE.
V2 · Historical network	History is no longer a single line. Concepts connect to people, places, texts, artifacts, disputes, industries and later applications. A Concept can have roots in one world, a breakthrough in another, and mass transmission somewhere else.

2. Editorial law: follow what survived

This is not intended to be a diplomatic survey in which every civilization receives equal space. Nor is the project trying to avoid looking Eurocentric, Sinocentric, Indic, Islamic, American, or anything else when the historical evidence happens to cluster strongly in one place. The network follows the mathematics.

The preferred historical lens is individual-centered and transmission-centered: who made an idea durable? Who formalized it, named it, gave it notation, published it, taught it, built an instrument around it, commercialized it, or made it part of an industry? Earlier precursors remain important - especially when priority is disputed - but they do not automatically displace the person or work through which the later mathematical tradition actually inherited the idea.

The rule is therefore not 'first person wins.' It is closer to: roots + breakthrough + transmission + industry + modern form.

This is explicitly a hobby/research framing. If Mathema later becomes a public school-facing educational product, the editorial choices should be reviewed again for age, curricular context, attribution standards and presentation.

3. Attribution is a network, not an inventor field

Role	Meaning
ROOTS	Earliest known precursor, evidence, or recognizable ancestor.
DISCOVERER	Strong claim to independent discovery of the mathematical result.
FORMALIZER	Turns an insight into a precise, reusable mathematical object or theory.
NOTATOR	Introduces notation that materially changes usability or transmission.
PUBLISHER / TRANSMITTER	Gets the idea into durable circulation: book, translation, school, journal, table, standard text.
POPULARIZER	Makes an idea broadly understood or routinely used.
INDUSTRIALIZER	Embeds the mathematics in a scalable instrument, profession, company, or technology.
DISPUTED / APPROPRIATED?	Records priority fights, borrowing, missing credit, independent rediscovery, or controversial transmission.
MODERN FORM	The form Mathema ultimately teaches and the later structure into which the idea grew.

The headline can privilege the person who changed the downstream world while the asterisk tells the messier priority story. Controversy is not a reason to remove a story; it is often the reason the story is memorable.

4. The seven Math-History Eras

These are not general-history periods. They are epochs in humanity's external mathematical mind: each is defined by a technology that changes what mathematics can store, transmit, execute, or search.

Era	Thesis	Artifacts	Mathematical shift
I · MARK	Mathematics can be stored.	Tally, token, clay, glyph, numeral.	Quantity escapes biological memory.
II · WRITE	Mathematics can be argued.	Scroll, codex, diagram, compass, straightedge.	Long procedures, diagrams and proofs can survive inspection.
III · ALGORITHM	Mathematics can be proceduralized.	Counting board, positional numerals, zero, tables, astrolabe, manuscript algorithms.	Reusable procedures become portable technologies of calculation.
IV · PRINT	Mathematics can spread.	Printing press, printed algebra, standardized notation, logarithm table, navigation chart, telescope.	Replication accelerates; notation and methods compete and standardize.
V · MECHANIZE	Mathematics can be embodied.	Slide rule, planimeter, mechanical calculator, engineering drawing, graph paper.	Mathematical operations become physical instruments and professional infrastructure.
VI · COMPUTE	Mathematics can execute itself.	Punch card, electronic calculator, mainframe, PC, spreadsheet, CAS, graphing calculator.	Machines perform procedures at scale; formulation matters more than hand execution.
VII · INTELLIGENCE	Mathematics can increasingly search itself.	Proof assistant, symbolic search, neural systems, AI reasoning tools.	Machines increasingly search representations, strategies, models, proofs and solutions.

5. Twelve mathematical Worlds

The Eras answer how mathematics was being done. The Worlds answer where particular mathematical ecosystems became unusually fertile. Worlds overlap in time and exchange ideas; they are not twelve isolated civilizations and not a ranking.

World	Signature	What comes alive
01 · MESOPOTAMIA	Clay-tablet computation	Place value, tables, equations, measurement, astronomy. Excellent MARK-era world: mathematics as administration and calculation.
02 · EGYPT / EASTERN MEDITERRANEAN	Survey, measure, record	Fractions, geometry, building, calendars, bureaucratic arithmetic; a strong physical world of rope, papyrus and monument.
03 · GREEK / HELLENISTIC	Proof and geometric structure	Euclid, Archimedes, Apollonius; theorem, proof, construction, conics, mathematical astronomy.
04 · CHINA	Algorithmic state mathematics	Counting rods, elimination, negative quantities, surveying, calendars, practical algorithms. The Nine Chapters became a durable computational canon.
05 · INDIA	Numeral and computational abstraction	Decimal place value, zero, algebraic rules, trigonometry, astronomical computation, later infinite-series work.
06 · BAGHDAD / ISLAMIC-PERSIAN WORLD	Translation -> algebra -> synthesis	House of Wisdom, al-Khwarizmi, algebra, algorithms, astronomy, trigonometry, translation and recombination of Greek/Indian traditions.
07 · ITALIAN COMMERCIAL / RENAISSANCE	Merchant math becomes print math	Fibonacci transmission, abacus schools, bookkeeping, perspective, cubic/quartic contests, printed arithmetic and algebra.
08 · ATLANTIC SCIENTIFIC REVOLUTION	Instrument + experiment + symbolic mathematics	Navigation, telescope, logarithms, analytic geometry, probability, calculus, mathematical physics.
09 · FRENCH ENLIGHTENMENT / REVOLUTIONARY	Mathematics becomes an institution	Analysis, probability, mechanics, descriptive geometry, metric standardization, grandes écoles and professional mathematical training.
10 · CENTRAL EUROPE / 19TH-CENTURY ABSTRACT	Rigor and structure	Gauss onward: number theory, non-Euclidean geometry, rigorous analysis, sets, abstract algebra and structural mathematics.
11 · BRITISH INDUSTRIAL / ENGINEERING	Mathematics as infrastructure	Surveying, navigation, actuarial work, statistics, engines, telegraphy, slide rules and engineering computation.
12 · AMERICAN / GLOBAL COMPUTATIONAL	Electronics -> software -> AI	Information theory, digital computers, numerical methods, operations research, computer graphics, machine learning, proof assistants and AI.

6. China can become a spectacular mathematical World

China is especially useful because its mathematical tradition looks different from the Greek proof-centered stereotype while still connecting directly to skills Mathera teaches. The Nine Chapters on the Mathematical Art is a practical problem collection involving surveying, trade, taxation, fractions, areas and equation-like procedures; later commentary by Liu Hui supplied sophisticated reasoning and approximation. It functioned as a major mathematical textbook tradition for roughly fifteen centuries.

Visually, this World has unusually strong objects: counting rods, counting boards, bureaucratic ledgers, land-survey diagrams, calendars, astronomical instruments, magic squares, printed mathematical manuals.

Concept links worth exploiting include: positional representation without a written zero on the counting board; negative numbers represented with rods; elimination methods that feel strikingly like later linear-system algorithms; iterative approximation of π ; surveying by similar triangles; and the tight historical coupling between mathematics, astronomy and calendrical administration.

This is a good example of the network principle. A Concept need not be assigned permanently to 'China.' A Chinese node can be one powerful milestone in a Concept whose roots or later industrial life occur elsewhere.

7. Transmission chains are more interesting than ownership

The network should actively show ideas crossing Worlds. Decimal positional numeration and zero are a model case: Indian mathematical traditions are central to the system that ultimately became modern Hindu-Arabic numeration; Islamic scholars including al-Khwarizmi transmitted and developed arithmetic and algebraic methods; Latin translations and figures such as Fibonacci became important European transmission points.

Likewise, al-Khwarizmi is compelling not merely because of a priority claim. His algebra became a durable textbook tradition, Latin translations influenced medieval European algebra, and his name ultimately supplied the word algorithm. That is exactly the kind of downstream survival Mathera wants to foreground.

The editorial question therefore becomes: Which node changed the downstream network? Earliest evidence is one node; the person whose book, notation or instrument conquered later practice may be another.

8. Artifacts: tools of math and things math built

Every Concept can connect to physical culture in two different ways.

TOOLS OF MATH - objects used to perform, store or transmit mathematics: clay tablet, abacus, astrolabe, compass, printing press, logarithm table, slide rule, planimeter, calculator, computer, proof assistant.

THINGS MATH BUILT - objects or systems that visibly embody mathematics: telescope, clock, bridge, railway, telegraph, television, aircraft, cryptography, GPS, computer graphics, machine learning.

This distinction lets a Concept card tell two stories at once: how humans worked with the mathematics, and what the mathematics eventually made possible.

9. People become headline nodes

The historical network should contain short, aggressively interesting story hooks attached to Concepts. The headline may be clickbait-like; the article underneath should be careful, sourced and explicit about uncertainty.

Person / story	Headline	Concept-network use
Kepler	THE ASTRONOMER WHO HAD TO DEFEND HIS MOTHER IN A WITCH TRIAL	Ellipses, planetary motion, mathematical astronomy, PRINT-era scientific networks.
Farnsworth	THE FARM BOY WHO TURNED AN IDEA INTO ELECTRONIC TELEVISION - THEN FOUGHT OVER THE PATENTS	Signals, scanning, electronics, THINGS MATH BUILT, industrialization and patent history.
Tesla	THE INVENTOR WHO TRIED TO BUILD A WORLD-SCALE WIRELESS SYSTEM	Alternating current, resonance, engineering mathematics, spectacular ambition and commercial failure.
Cardano / Tartaglia / Ferrari	THE SECRET CUBIC SOLUTION, THE BROKEN PROMISE, AND THE PUBLIC MATH FIGHT	Polynomial equations, publication, priority and Renaissance mathematical competition.
Newton / Leibniz	THE CALCULUS WAR	Independent discovery, notation, institutions, priority, publication and the politics of mathematical survival.
Sophie Germain	SHE USED A MALE NAME TO SEND HER MATHEMATICS	Number theory, elasticity, access to institutions, correspondence networks.
Shannon	THE FATHER OF INFORMATION THEORY JUGGLED ON A UNICYCLE AT BELL LABS	Information, computation, probability, engineering and the COMPUTE world.
Galois	THE YOUNG MATHEMATICIAN WHO DIED AFTER A DUEL	Algebraic structure, manuscripts, posthumous transmission and mathematical myth-making.

Useful story tags: FEUD, DOOM, INVENTION, EUREKA, TRIAL, SECRET, EMPIRE, MYTH?, OBSESSION, REBEL. A disputed anecdote can remain if it is clearly labeled as tradition, disputed, or myth rather than silently presented as fact.

10. The data model

Each of the 100₁₂ Concepts can now be represented as a small historical network rather than a single dated card.

Field	Purpose
Concept	e.g. Multiplication
Mathera encounters	Branches and Skills that train it; learner depth and retained strength.
Technology Era	MARK / WRITE / ALGORITHM / PRINT / MECHANIZE / COMPUTE / INTELLIGENCE.
World nodes	One or more of the twelve mathematical ecosystems.
Roots	Earliest known evidence or precursor.
Breakthrough	Person/work that gives the idea a recognizable powerful form.
Transmission	Text, translation, notation, school, instrument or institution that makes it durable.
Industry	Profession, company, artifact or mass technology that scales it.
Modern form	What the Concept becomes in later mathematics.
People / stories	Biographical hooks, disputes, triumphs, disasters, oddities.
Artifacts	Tools of Math + Things Math Built.
Confidence	Exact / approximate / disputed / conventional attribution, with source notes.

11. What the learner ultimately sees

A learner opens a Concept such as Multiplication. Instead of a sterile progress bar, the page can show the learner's own encounters alongside a human technology tree: early recorded arithmetic, geometric interpretations, written algorithms, printed symbolic algebra, mechanical computation, matrices and digital computation.

Clicking sideways opens Worlds and people. A Chinese elimination procedure can sit beside later linear algebra. An Indian numeral milestone can connect through Baghdad to a Latin transmission node. A Renaissance printing node can explain why one notation conquered another. A slide rule can show logarithms becoming a physical machine. Television can appear downstream from Fourier/signal mathematics. AI can appear as the newest layer in the automation of mathematical search.

The point is not to force civilization into a neat procession. It is to let the learner wander through a network of mathematical inheritance.

12. Research notes supporting the World/network framing

The following checks support the direction of the design. They are not a complete historiography.

China. The Nine Chapters is described by MacTutor as a practical handbook of 246 problems involving engineering, surveying, trade and taxation; it became foundational to Chinese mathematics and remained a textbook tradition for many centuries. Liu Hui's 263 CE commentary added sophisticated mathematical reasoning and iterative approximation. Chinese counting rods also provided a positional computational system in which an empty place on the board could serve the role of zero.

India. Historical scholarship credits Indian traditions with crucial development of decimal positional numeration and zero; Aryabhata's school linked mathematics closely with astronomy and trigonometry, and the Kerala tradition later developed notable infinite-series results.

Baghdad / Islamic mathematics. Al-Khwarizmi's work is an unusually good transmission node: algebra becomes a systematic practical theory; his arithmetic work describes Hindu numeration; Latin translations became influential in medieval Europe; and his name is the source of the word 'algorithm.'

Network rather than relay race. The historical record does not support a simple one-civilization-after-another chain. Ideas moved, were independently rediscovered, translated, reformulated and sometimes lost and revived. That is precisely why the World + Era + Person + Artifact network is preferable to a single timeline.

Research anchors used for this revision

MacTutor History of Mathematics: History overview; Chinese overview; Nine Chapters on the Mathematical Art; Chinese numerals; Indian mathematics; Zero; Arabic mathematics; al-Khwarizmi biography and related historical essays. These should be supplemented with primary/academic sources when individual Concept cards receive exact dates or contested priority claims.

13. Final settled architecture

LAYER 1 - JOURNEY

Count -> Operate -> Relate -> Solve -> Prove -> Change -> Space

The learner's ordered path through Branches, Skills and Levels.

LAYER 2 - CAPABILITY

100₁₂ Concepts

The recurring mathematical powers being strengthened across the entire course.

LAYER 3 - TECHNOLOGY ERA

MARK -> WRITE -> ALGORITHM -> PRINT -> MECHANIZE -> COMPUTE -> INTELLIGENCE

How humanity externalized, transmitted and automated mathematical thought.

LAYER 4 - WORLDS

Twelve overlapping mathematical ecosystems

Where particular combinations of institutions, problems, people and tools became unusually fertile.

LAYER 5 - HUMAN NETWORK

People -> texts -> disputes -> instruments -> industries -> artifacts -> later technologies

The stories that make the mathematics memorable.

**One system describes the development of a mathematical mind.
The other describes the development of humanity's external
mathematical mind. The network connects them.**